# FIFA 2019

## Player Data Analysis

A complete data science project using 7 analytical techniques to find real insights inside 18,207 football players.

**18,207**

players analysed

**88**

columns of data

**7**

advanced techniques

**5**

key business insights

# The Scenario — Why Are We Doing This?

*Imagine you work as a data analyst for a football club. Your boss gives you one file.*

## The Mission

Your club has a budget to sign players.

You have FIFA data on 18,207 players.

Your boss asks:

- Who is worth signing?
- Who is overpaid right now?
- Where is the hidden value?
- Which young players will grow?

Data science is how you answer these.

## What is in the dataset?

Player name, age, nationality, club

Overall rating (1-99) and Potential rating

Weekly wage and market value (in euros)

Position (GK, CB, CM, ST …) and preferred foot

60+ individual skill stats (passing, speed, shooting…)

**18,207**
players

**€226M**
highest value

**€565K**
highest wage/wk

**94**
highest Overall rating

# Step 1 — Cleaning the Data

*Real data is messy. Before any analysis, we have to fix it.*

**PROBLEM**

"€110.5M" stored as text

→

**SOLUTION**

Converted to number: 110,500,000

**PROBLEM**

"5'7" (feet & inches)

→

**SOLUTION**

Converted to cm: 170.2 cm

**PROBLEM**

27 position codes: GK, CB, CDM…

→

**SOLUTION**

Grouped into 4: GK, Defender, Mid, Forward

**PROBLEM**

Players with no wage listed

→

**SOLUTION**

Flagged and kept — not deleted

# Technique 7 — Missing Data Patterns

*Think of it like this: if someone skips the salary box on a job form, that is probably not random. It tells you something about them.*

## What We Found: Release Clause is Missing for 8.6% of Players

### Release Clause PRESENT

16,643 players

**Avg Overall: 66.2**

**Avg Value: €2.44M**

**Avg Wage: €9.6K/wk**

### Release Clause MISSING

1,564 players

**Avg Overall: 67.3**

**Avg Value: €2.14M**

**Avg Wage: €11.3K/wk**

INSIGHT: Players with no release clause earn more per week — but are worth less overall. They are on short-term deals. The club is not investing in their future.

# Technique 6 — Ratio and Derived Metrics

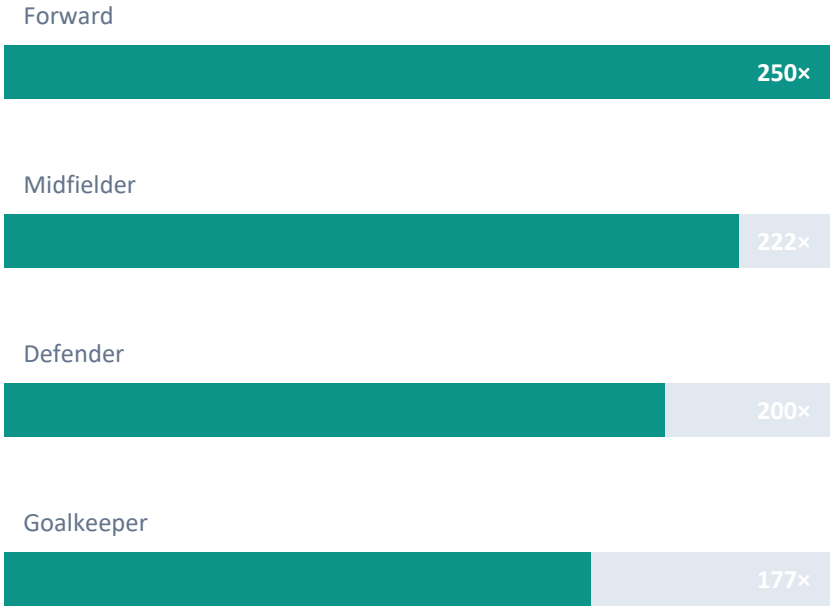
*We create new columns the data did not have, to measure EFFICIENCY — not just size.*

## Potential Gap = Potential − Overall

*How much room does this player have to grow?*

## Value-to-Wage = Market Value ÷ Weekly Wage

*How much market value does the club get per euro of wage?*



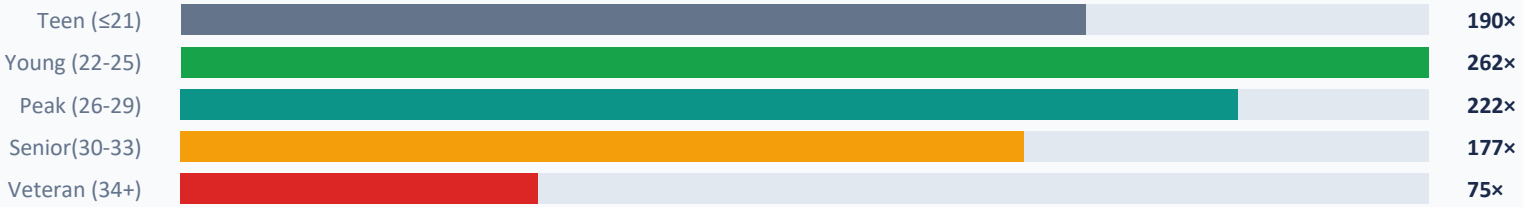

INSIGHT: Forwards give clubs the most market value per euro of salary. Young gems (low wage, high potential gap) are the best investment of all.

# Technique 4 — Cohort Analysis

*Think of it like comparing different departments in a company. Averaging them all together hides how different they really are.*

| Age Group | Players | Avg Overall | Avg Value | Avg Wage/wk | Value-to-Wage |
|---|---|---|---|---|---|
| Teen (≤21) | 4,738 | 60.8 | €1.16M | €4.2K | 190× |
| **Young (22-25)** | **5,328** | **66.7** | **€2.79M** | **€9.4K** | **262× ★ BEST** |
| Peak (26-29) | 4,593 | 69.0 | €3.32M | €13.6K | 222× |
| Senior(30-33) | 2,599 | 69.7 | €2.86M | €14.3K | 177× |
| Veteran(34+) | 889 | 68.7 | €0.93M | €8.4K | 75× |

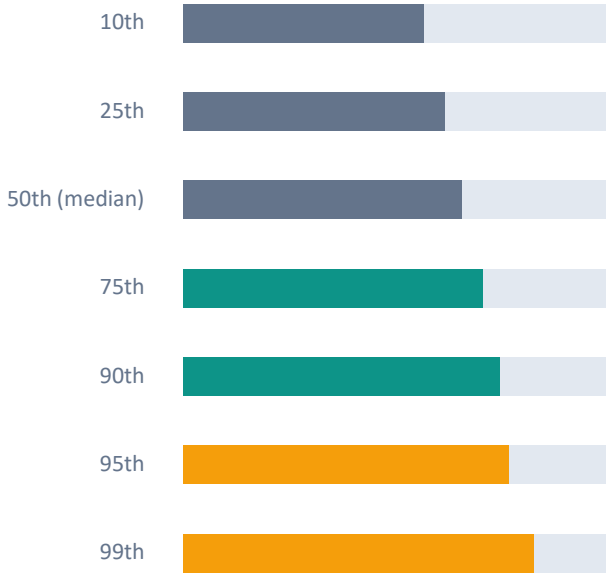
**Value-to-Wage by Age Group (higher = better deal for the club)**



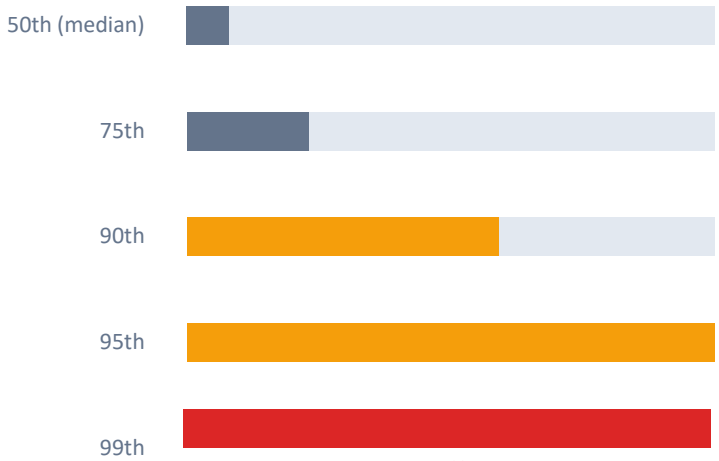
# Technique 3 — Percentile Analysis

*If Messi earns €565K/week and 9 reserve players earn €1K/week — the average wage is €57K. That describes nobody. The median is €1K. That is the truth.*

## Overall Rating Percentiles




## Wage Inequality




**Top 5% of earners consume 41% of all wages**

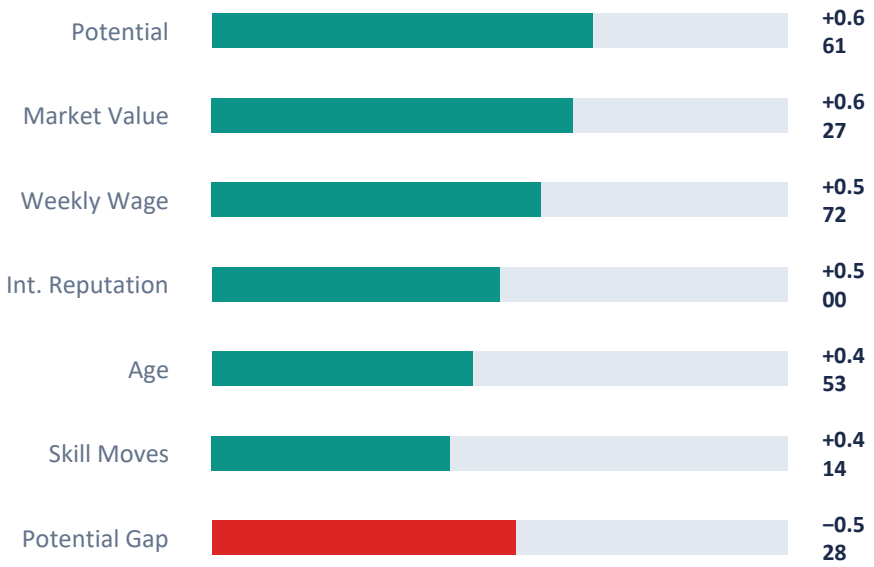
The median player earns €3K/week. That is 13× less than the 95th percentile player.

INSIGHT: Football wages follow extreme inequality. A budget club cannot compete on wages — it must compete on smarter analytics.

# Technique 2 — Correlation Analysis

r = +1 means: when X goes up, Y always goes up too. r = 0 means: no link at all. r = −1 means: opposite directions.

## What drives Overall rating?




## The Youth Premium

We looked at players with the SAME skill (Overall 83-87) but different ages.

**€40M**
Players aged ≤25

**€23M**
Players aged ≥30

**1.7×**
higher price for youth — at identical skill

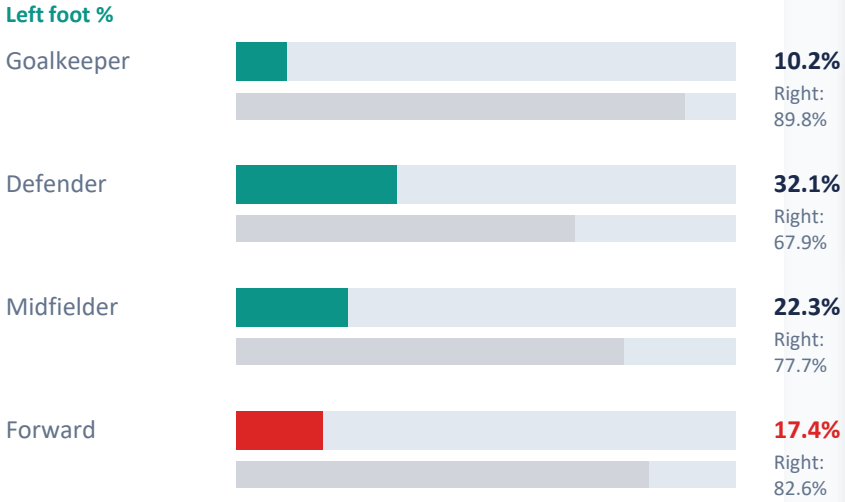
*Clubs pay for future years — not just today's ability.*

INSIGHT: Age drives value independently of skill. A 24-year-old and a 33-year-old with the same rating are NOT the same price.

# Technique 1 — Cross-Tabulation Analysis

*Like a pivot table in Excel. Instead of asking 'how many forwards?' — we ask 'how many LEFT-FOOTED forwards?' That extra layer reveals something new.*

## Preferred Foot by Position




## The Scarcity Premium

*Among forwards with the same skill level:*

**€1.50M**
Left-footed forwards

**€1.20M**
Right-footed forwards

**+25%**

higher median market value for the rare left foot

*Only 17% of forwards are left-footed — but defenders train all week against right-footers. A left-footer attacks from unexpected angles.*

INSIGHT: Scarcity creates price premiums. Left-footed youth forwards are undervalued before they become famous. Scout them early.

# Technique 5 — Outlier Investigation

*A hospital sees a heart rate of 240 BPM. Delete it as an error? No — that patient might be in cardiac arrest. Same logic in our data.*

## IQR Method

1. Q1 = 25th percentile of wages = €3,000 /week
2. Q3 = 75th percentile of wages = €9,000 /week
3. IQR = Q3 − Q1 = €9,000 − €3,000 = €6,000
4. Upper limit = Q3 + (1.5 × IQR) = €9,000 + €9,000 = €18,000
5. **Any player earning > €18,000/week = WAGE OUTLIER**

## Are Outliers Actually Better?

| | Normal players | Wage outliers |
|---|---|---|
| Avg Overall | 64.9 | 76.9 |
| Avg Potential | 68.5 | 80.1 |
| Int. Reputation | 1.1 | 2.8 |
| % of all players | 88.8% | 11.2% |
| % of all wages | 40.9% | 59.1% |

**11.2%**
of players are outliers

**59.1%**
of wages go to outliers

**12 pts**
higher avg Overall

INSIGHT: Wage outliers are NOT errors. They are the elite VIP segment — genuinely better players who command top wages because they deserve them.

# The 5 Key Insights from Our Analysis

## 1 Wage inequality is extreme

T3 + T5

Top 5% of earners take 41% of all wages. Median player earns €3K/wk — 13× less than the 95th percentile.

## 2 Youth costs more at the same skill

T2 + T4

Same skill (83-87 Overall): a 24-year-old is worth 1.7× more than a 33-year-old. Clubs pay for future years.

## 3 Left-footed forwards earn a 25% premium

T1

Only 17% of forwards are left-footed. Rarity creates a tactical advantage — and a price premium.

## 4 Missing release clause = contract signal

T7

These players earn well short-term but have lower long-term value. Missing data was NOT random.

## 5 Young gems are the best ROI

T6

Age ≤23, Potential Gap ≥8, wage <€30K/wk. These players give the highest value-to-wage ratio of all.

# What Should a Football Club Actually Do?

*Data science is only useful if it leads to real decisions.*

### A — Target the 22-25 age bracket

Best value-to-wage ratio (262×). Proven ability but wages not yet inflated by reputation.

### B — Use the Young Gems filter

Age ≤23, Potential Gap ≥8, Wage <€30K/week. Find stars before they become expensive.

### C — Scout left-footed youth forwards

Only 17% of forwards are left-footed. They create unpredictable angles and carry a 25% value premium.

### D — Sell veteran stars before age 30

A player's market value drops sharply after 30. Sell while value is still high. Reinvest in youth.

### E — Treat missing RC as a data signal

Players without a release clause are on short contracts. Use this flag in any predictive model — do not delete these rows.

### F — Do not compete on wages alone

The top 5% take 41% of wages. Budget clubs cannot win that fight. Win with smarter analytics instead.

# Summary — The 7 Techniques in One View

| # | Technique | The Question It Asks | What We Found |
|---|---|---|---|
| T7 | **Missing Data Patterns** | Is missing data random — or a signal? | **Release Clause absence = expiring contract flag** |
| T6 | **Ratio & Derived Metrics** | What efficiency columns can we create? | **Potential Gap, Value-to-Wage ratio** |
| T4 | **Cohort Analysis** | Do different age groups behave differently? | **Young (22-25) has best ROI: 262× ratio** |
| T3 | **Percentile Analysis** | What is the real distribution of the data? | **Top 5% earners take 41% of all wages** |
| T2 | **Correlation Analysis** | What variables are linked — and how strongly? | **Youth premium: 1.7× value at same skill** |
| T1 | **Cross-Tabulation** | What patterns appear when we cross two categories? | **Left-footed forwards earn 25% more** |
| T5 | **Outlier Investigation** | Are extreme values mistakes or real findings? | **Wage outliers = VIP players, not errors** |

*Each technique is a different lens. Alone they are interesting. Together they are powerful.*

**Data science is not about calculating numbers.**

It is about asking better questions.

# Thank You

FIFA 2019 · Player Data Analysis Project

- 18,207 players analysed
- 7 analytical techniques applied
- 5 clear business insights found
- Every insight backed by real data

*The 7 techniques are 7 different ways of asking: what is actually happening here?*